Phase 1: Project Scoping and Task Distribution

Your target is 300 high-quality, perfectly constrained prompts distributed across four languages and seven core tasks and each prompt will have 1 to 10 individual tasks.

Language Distribution Matrix

- **English:** 101 prompts (25%)
- **Chinese:** 75 prompts (25%)
- **Arabic:** 62 prompts (25%)
- **Hindi:** 62 prompts (25%)

Core Task Categories

- Classification
- Information Extraction
- Text Generation
- Dialogue System
- Reasoning and Logic
- Language Style
- Programming (Evaluation and Verification embedded across tasks)

Constraint Scaling Logic

Every task within each prompt will contain between 1 and 5 distinct, atomic constraints. The target word count will scale dynamically based on the number of constraints N embedded within the prompt. Token count will act as a secondary internal metric.

- **Low Density (1-2 constraints):** Smaller word count limit.
- **Medium Density (3-4 constraints):** Moderate word count limit.
- **High Density (5 constraints):** Larger word count limit to allow the model sufficient space to satisfy all requirements without artificial clipping.

Phase 2: The Data Generation and Localization Pipeline

This phase ensures all constraints are simultaneously achievable and culturally resonant.

Step 1: English Seed Generation

Generate a set of base tasks in English using an LLM. After producing the full list of tasks (N), perform a second pass where 1–5 constraints are added to each task. These constraints should be drawn from the categories of Content, Situation, Style, Format, and Length.

Step 2: Domain-Specific Adjustments

- **Programming Tasks (Arabic/Hindi):** Explicitly instruct the model to write the functional syntax, code, and variables in English, but output all docstrings, inline comments, and explanatory summaries in the target language (Arabic or Hindi).
- **Dialogue Tasks:** Explicitly state that length and format constraints apply to the entire generated conversation collectively, not individual turns.

Step 3: Translation and Cultural Localization

Translate the English seed prompts into Chinese, Arabic, and Hindi. Replace English-centric cultural anchors with localized equivalents (e.g., swapping a reference to "YouTube" for "Hotstar" in Hindi). Annotate these prompts with Cultural Accessibility Labels.

Step 4: Back-Translation Verification

Use Google Translate and LLMs to back-translate the localized prompts into English. Verify that the semantic intent and the atomic constraints remain perfectly intact. Discard or rewrite any prompt where the logic breaks down during translation.

Phase 3: Instruction-Level Validation

Before a prompt enters the final benchmark dataset, it must pass a rigorous, LLM-driven extraction and feasibility check.

Automated Requirement Extraction

Prompt an LLM to extract requirements from the generated instructions based on four pillars:

- **Explicitness:** Is it clearly stated?
- **Completeness:** Are all constraints accounted for?
- **Atomicity:** Does this specific extraction address a single, measurable aspect?
- **Categorization:** Is it correctly labeled (e.g., Content vs. Style)?

Feasibility and Conflict Resolution

Run an automated logical consistency check to ensure no contradictory constraints exist (e.g., demanding a 50-word limit while simultaneously requiring a comprehensive 5-column Markdown table). Remove redundant, infeasible, or language-dependent constraints (like specific character counts or alliteration).

Phase 4: Hybrid Evaluation and Scoring Mechanism

When testing models against the 300 prompts, the evaluation will be split into objective scripts and subjective LLM judging to prevent bias.

Deterministic Script Evaluation (Objective)

Use Python scripts and Regex to evaluate rigid constraints.

- **Format Constraints:** Verify Markdown tables, JSON structures, or numbering lists.
- **Length Constraints:** Calculate exactly how closely the model adhered to the dynamic word count.
- **Length Deviation:** This measures magnitude and direction of failure using the formula:
- delta L = L_pred - L_gt / L_gt
- **Length Score:** Aggregates the deviation into a robust performance metric.

LLM-as-a-Judge Evaluation (Subjective)

Deploy a strong, independent LLM to act as the judge for nuanced constraints using the extracted atomic requirements.

- **Style and Situation:** Verify Tone, Emotion, and Role-Playing adherence.
- **Content:** Verify Privacy exclusions or Theme inclusions.

Phase 5: Post-Evaluation Analysis

Once the models have been scored, calculate your final benchmarking metrics to analyze the results.

Core Metrics

- **RFR (Requirement Following Ratio):** The proportion of satisfied requirements over all instructions for a specific language.
- **IFR (Instruction Following Ratio):** The proportion of instructions where *all* requirements were satisfied.

Cultural and Linguistic Gap Analysis

Cross-reference the RFR and IFR scores with the Cultural Accessibility Labels. This will isolate whether a model failed due to the raw compositional complexity of the 5 constraints, or because it lacked the cultural context required for the localized Arabic or Hindi references.

https://aclanthology.org/2025.emnlp-main.1059.pdf
https://github.com/Hope-Rita/EIFBench
https://github.com/Tongyi-CCAI/Complex-IF

https://arxiv.org/pdf/2503.07539
https://github.com/zhenyuli801/XIFBench

https://arxiv.org/pdf/2505.16234